

FDM-X300 Industrial 3D Printer

User & Maintenance Manual

Firmware Version: 3.4.1 **Manual Version:** 2.1 **Last Updated:** February 2024

Welcome to the FDM-X300

The FDM-X300 is a high-performance industrial fused deposition modeling (FDM) 3D printer built for continuous production environments. Designed for reliability and precision, the FDM-X300 supports a wide range of engineering-grade thermoplastic filaments and delivers consistent part quality across long unattended print runs.

This manual covers technical specifications, normal operating parameters, error diagnosis and resolution, maintenance procedures, and safety guidelines. Please read this manual fully before operating the machine.

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Section 1 — Technical Specifications

Build Volume

- X axis: 300 mm
- Y axis: 300 mm
- Z axis: 400 mm

Print System

- Print technology: Fused Deposition Modeling (FDM)
- Number of extruders: 2 (primary + support)

- Nozzle diameter (standard): 0.4 mm
- Available nozzle diameters: 0.25 mm, 0.4 mm, 0.6 mm, 0.8 mm
- Max nozzle temperature: 300°C
- Nozzle material: Hardened steel (standard), Brass (optional)
- Filament diameter: 1.75 mm
- Max volumetric flow rate: 24 mm³/s

Motion System

- XY motion: CoreXY belt drive
- Z motion: Dual lead screw
- Max print speed: 180 mm/s
- Positioning resolution: 0.01 mm

Build Platform

- Heated bed: Yes
- Max bed temperature: 120°C
- Bed leveling: Automatic (9-point inductive probe)
- Bed surface: Removable PEI spring steel sheet

Enclosure

- Fully enclosed build chamber
- Active chamber heating: up to 70°C
- HEPA + activated carbon filtration: Yes
- Chamber temperature sensor: Yes

Connectivity

- USB, Ethernet, Wi-Fi
- Touchscreen display: 7 inch colour

Power

- Input voltage: 220V / 50Hz single phase
- Max power draw: 2,000W
- UPS recommended: Yes

Physical

- Dimensions (W×D×H): 620 × 580 × 780 mm
- Weight: 48 kg

Supported Materials PLA, PETG, ABS, ASA, TPU, Nylon (PA12), PC, PLA-CF, PETG-CF, ABS-CF

Section 2 — Normal Operating Parameters

During a healthy print run, the following parameters should fall within the ranges below. Values outside these ranges during printing indicate a potential issue.

Parameter	Expected Range
Nozzle temperature (during print)	Set point $\pm 3^{\circ}\text{C}$
Bed temperature (during print)	Set point $\pm 5^{\circ}\text{C}$
Chamber temperature (enclosed prints)	Set point $\pm 8^{\circ}\text{C}$
Filament feed rate deviation	Within $\pm 5\%$ of set value
Print head position error	Below 0.05 mm
Cooling fan speed (part cooling)	30% – 100% depending on material
HEPA filter pressure drop	Below 120 Pa
Z offset (first layer)	–0.1 mm to –0.3 mm from probe trigger

Section 3 — Error Codes and Troubleshooting

ERR-01 — Nozzle Temperature Runaway

What this means: The nozzle temperature has either dropped more than 15°C below the set point during printing, or has risen more than 20°C above the set point unexpectedly. Both conditions trigger an immediate print pause and heater shutdown to prevent damage or fire risk.

Why it happens:

- The heater cartridge inside the hotend has failed
- The thermistor (temperature sensor) inside the hotend has failed or has a loose connection
- A draft or airflow is cooling the nozzle during printing (enclosure door left open)
- Firmware PID tuning values are incorrect after a nozzle change

How to fix it:

If temperature dropped below set point:

1. Go to Settings → Diagnostics → Heater Test. Run the heater test for the affected extruder. If the nozzle temperature does not rise within 30 seconds, the heater cartridge has failed and must be replaced.

2. If the heater test shows temperature rising normally, check that the enclosure door is fully closed and latched — drafts cause rapid temperature loss at the nozzle.
3. If the door is closed and the heater is functioning, the thermistor connection may be intermittent. Power off the printer, open the printhead access panel (2 screws on the left side of the gantry), and reseal the thermistor connector.

If temperature rose above set point:

1. This indicates the thermistor is reading incorrectly low, causing the heater to keep running. Power off the printer immediately.
 2. The thermistor must be replaced before restarting. Contact your maintenance technician.
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ERR-02 — Bed Leveling Failed

What this means: The automatic bed leveling routine (9-point probe sequence) has failed to complete successfully. One or more probe points returned a reading outside the acceptable flatness range, or the probe did not trigger during the sequence.

Why it happens:

- The PEI sheet is not fully seated on the magnetic bed surface
- Print residue or debris on the bed surface is interfering with probe readings
- The inductive probe is too far from the bed surface — Z offset needs adjustment
- The bed surface has significant warp from thermal cycling

How to fix it:

1. Remove the PEI spring steel sheet from the bed. Clean the magnetic surface beneath it with a dry cloth. Reattach the PEI sheet — confirm it lies flat with no lifted corners.
 2. Clean the top surface of the PEI sheet with isopropyl alcohol (IPA) — 90% or higher. Remove all grease, residue, and fingerprints.
 3. Go to Settings → Bed Leveling → Manual Z Offset. Lower the Z offset by 0.05 mm increments until the probe triggers reliably during a test probe.
 4. Re-run the auto bed leveling routine. If it fails again and the bed surface has visible high and low spots, the PEI sheet may be warped and needs replacement.
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ERR-03 — Filament Runout or Feed Failure

What this means: The filament runout sensor has detected that filament is no longer present in the feed path, or the extruder motor has stalled due to a blockage or excessive back pressure.

Why it happens:

- Filament spool has run out
- Filament has snapped inside the bowden tube or at the extruder
- Partial nozzle clog causing back pressure that stalls the extruder motor
- Filament has tangled on the spool

How to fix it:

If spool is empty:

1. The printer pauses automatically and prompts filament change on the touchscreen. Follow the on-screen filament change procedure. Load the new spool and confirm filament is extruding cleanly before resuming.

If filament is present but feed has failed:

1. Open the filament access panel on the side of the machine. Check whether the filament is tangled on the spool — untangle and ensure the spool rotates freely on its holder.
2. Check the bowden tube connection at the extruder end — press the collet clip and reseal the tube if loose.
3. If the filament is present and the path is clear but the extruder is still not feeding, a partial nozzle clog is likely. Perform a cold pull procedure: heat the nozzle to print temperature, then cool to 90°C, then manually pull the filament straight out. Repeat 2 to 3 times until the pulled filament tip comes out clean with no dark residue.
4. If cold pull does not clear the clog, heat the nozzle to 260°C and perform a manual purge of 50 mm of filament. If flow is still restricted, the nozzle must be replaced.

ERR-04 — Chamber Temperature Not Reached

What this means: The build chamber has not reached the set chamber temperature within 20 minutes of the heat-up command. Print has not started.

Why it happens:

- Enclosure door or top panel is not fully closed
- Chamber heater element has failed
- Chamber temperature sensor fault

How to fix it:

1. Check that all enclosure panels are fully closed and latched — even a small gap significantly slows chamber heat-up.
2. Go to Settings → Diagnostics → Chamber Heater Test. If the temperature does not rise during the test, the chamber heater element has failed. Contact your maintenance technician.
3. If the heater test shows temperature rising but the sensor reading is not changing, the chamber sensor is faulty. Contact your maintenance technician.

4. If the chamber is heating but slowly, check that the HEPA filter is not clogged — a blocked filter restricts airflow circulation inside the chamber. Check filter status in Settings → Maintenance → Filter Status.
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ERR-05 — Print Head Position Error (Axis Stall)

What this means: The motion system has detected that the print head did not reach its commanded position. This indicates the CoreXY belt system has stalled, skipped, or lost synchronisation.

Why it happens:

- Belt tension too low — belt slipping on pulley
- Print head physically obstructed — foreign object or print failure causing a collision
- A layer of the print has warped and lifted off the bed, causing the print head to collide with the part

How to fix it:

1. Cancel the current print. Home the print head using the touchscreen home command and observe whether the motion is smooth and without unusual noise.
 2. If the head moves smoothly after homing, the stall was likely caused by a print collision (warped part). Check bed adhesion settings and first layer parameters before reprinting.
 3. If motion is rough or noisy during homing, check belt tension: open the top panel of the printer. The CoreXY belts should be taut with no visible sag. If a belt is loose, tension it using the belt tensioner screws located at the rear of the gantry — refer to the tensioning guide in Appendix B.
 4. Check both belt paths for foreign objects, debris, or filament strands caught in the pulley teeth.
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ERR-06 — HEPA Filter Replacement Required

What this means: The measured pressure drop across the HEPA filter has exceeded 120 Pa, indicating the filter is saturated and airflow through the enclosure is restricted. This is a maintenance notification, not a fault — the printer continues operating but filter replacement is required soon.

Why it happens:

- Normal filter saturation from accumulated particulates during printing
- Accelerated clogging when printing high-emission materials (ABS, CF-filled filaments)

How to fix it:

1. This alert can be acknowledged and printing continued for up to 10 additional print hours before the filter must be replaced.
 2. To replace the filter: power off the printer. Open the rear access panel (4 thumb screws). Slide out the filter cartridge. Insert the new HEPA + activated carbon cartridge — part number FDM-X300-FL-001. Close the rear panel.
 3. After replacement, reset the filter counter in Settings → Maintenance → Reset Filter Counter.
 4. Do not operate the printer with a saturated filter when printing ABS, PC, or carbon fibre filled materials — these materials produce fine particles and fumes that require effective filtration.
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ERR-07 — Extruder Motor Overcurrent

What this means: The extruder motor current has exceeded the safe operating threshold. The extruder has been stopped to prevent motor damage.

Why it happens:

- Severe nozzle blockage creating excessive back pressure
- Filament has fused inside the hotend cold zone (heat creep)
- Extruder gear teeth worn — slipping and drawing high current intermittently

How to fix it:

1. Power off the printer and allow the hotend to cool fully to room temperature before handling.
 2. Once cooled, manually attempt to move the filament through the extruder by hand. Resistance indicates a blockage in the cold zone (heat creep) or the nozzle.
 3. Heat the nozzle to 260°C and attempt a manual purge. If material does not flow, the hotend is severely blocked and must be disassembled. Contact your maintenance technician.
 4. If purging restores flow, check the part cooling fan — heat creep is caused by insufficient cooling of the cold zone. The part cooling fan must run at minimum 30% during printing. Check fan operation in Settings → Diagnostics → Fan Test.
 5. Inspect the extruder gear teeth for wear — the teeth should be sharp and clearly defined. If worn smooth, replace the extruder gear assembly (part number FDM-X300-EG-003).
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Section 4 — Maintenance Schedule

After Every Print

- Remove completed print and clean the PEI sheet with IPA
- Check nozzle tip for material residue — wipe with a heat-resistant cloth while hot

- Inspect first layer adhesion quality — if poor, re-run bed leveling before next print

Weekly

- Check HEPA filter status in Settings → Maintenance → Filter Status
- Inspect bowden tubes for cracks, discolouration, or kinks — replace if damaged
- Check belt tension — belts must be taut with no sag
- Clean the build chamber interior — wipe walls with a dry cloth to remove dust and filament particles
- Lubricate Z-axis lead screws with a small amount of PTFE-based lubricant

Monthly

- Inspect extruder gear teeth for wear
- Check all hotend connections — thermistor and heater cartridge connectors must be firmly seated
- Verify Z offset calibration with a test print first layer
- Clean the inductive probe tip with IPA — contamination can cause false readings
- Inspect all cable chains on the gantry for cracked or broken links

Every 500 Print Hours

- Replace HEPA filter cartridge regardless of pressure drop reading
- Replace nozzle (standard 0.4 mm) — earlier if print quality has degraded
- Full hotend disassembly and inspection — check for wear at the heat break
- Lubricate all linear guide rails with appropriate machine oil

Section 5 — Safety Guidelines

- Do not leave the printer unattended during the first layer of a print — most print failures and potential hazards occur in the first 10 minutes
- Do not open the enclosure during printing of high-temperature materials (ABS, PC, Nylon) — chamber temperature and fume containment will be compromised
- Always operate the HEPA filtration system when printing ABS, ASA, PC, or carbon-fibre filled materials — these emit ultrafine particles and volatile compounds
- Do not touch the nozzle or bed surface during or immediately after printing — nozzle temperature can exceed 280°C and bed temperature can exceed 110°C
- In the event of a filament fire (rare but possible with severely clogged hotends) — press the emergency stop button on the rear of the machine and use a CO₂ fire extinguisher. Do not use water.
- Ensure the printer is placed on a stable, non-flammable surface with at least 200 mm clearance on all sides for ventilation
- Do not modify firmware settings outside of the parameters documented in this manual without consulting a qualified technician

Section 6 — Spare Parts Reference

Part Name	Part Number	Recommended Stock
HEPA + carbon filter cartridge	FDM-X300-FL-001	2 units
Standard nozzle 0.4 mm (hardened steel)	FDM-X300-NZ-002	4 units
Extruder gear assembly	FDM-X300-EG-003	1 unit
Heater cartridge (hotend)	FDM-X300-HC-004	2 units
Thermistor (hotend)	FDM-X300-TH-005	2 units
PEI spring steel sheet	FDM-X300-PS-006	1 unit
Bowden tube (per metre)	FDM-X300-BT-007	1 metre
CoreXY belt set	FDM-X300-BL-008	1 set
Chamber temperature sensor	FDM-X300-CS-009	1 unit

FDM-X300 User & Maintenance Manual v2.1 For technical support please contact your equipment maintenance team. Keep this manual accessible at the machine workstation at all times.